

46 YEARS OF EXCELLENCE

NARAYANA
IIT - JEE ACADEMY - INDIA

Sec: SR STAR - A & APEX, ALL IITS **DT: 18/01/2026**
Time: 3 Hrs **GTM - 18** **Max. Marks: 300**

PHYSICS **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

1. Regarding self inductance

i) The self inductance of coil depends on current flowing in the coil

ii) The self inductance of coil depends on magnitude of rate of change of current

iii) The self-inductance of coil depends on the geometrical shape and size of coil

iv) The self inductance depends on the magnetic permeability of the medium.

Choose the correct answer from the option given below

A) i, iii, iv only B) ii, iii, iv only

C) iii, iv only D) i, ii, iii, iv only

2. A solid wooden cylinder of height h is partially floating in water. The cylinder in depressed slightly and released to execute simple harmonic motion. The time period of oscillation is $y\pi \times 10^{-2}$ sec. What the value of y ? ($g = 10 m/s^2$, relative density of wood =0.36 and height of cylinder $h = 10 cm$)

A) 20 B) 12 C) 6 D) 25

For More Join: @IITJEE_Advanc

3. Given below are two statements

Statement-I: Viscosity of liquid decreases with increase in temperature.

Statement-II: Viscosity of gases increases with increase in temperature.

On the light of above statements, choose the correct answer from the options given below

A) Statement – I is false but statement II is true

B) Statement – I is true but statement II is false

C) Both statement – I and Statement – II are true

D) Both statement – I and Statement – II are false

4. A subatomic particle of mass $2.21\times10^{-30}kg$ is moving with speed $3\times10^{6}m/s$, under the matter wave consideration, the particle will behave closely like $\left(h=6.63\times10^{-34}J.s\right)$

A) Infra-red radiation B) Visible radiation

C) X-rays D) Gamma rays

5. One end of a long horizontal cylindrical glass rod $\left(\mu=3/2\right)$ of radius 5.0cm is rounded in the shape of a hemisphere. An object 0.5mm high is placed perpendicular to the axis of rod at a distance of 20.0cm from the rounded edge. The nature and size of image will be (object is in air and observer is in the cylindrical medium)

A) 0.5 mm, inverted B) 0.5 mm, erect

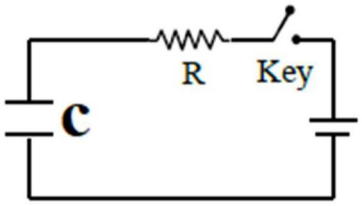
C) 0.75 mm, inverted D) 0.75 mm, erect

6. An alpha particle and a proton are accelerated through same potential difference starting from rest. Then the ratio of their de Broglie wave length is

A) 1 B) $\frac{1}{2\sqrt{2}}$ C) $\frac{1}{4}$ D) $\frac{1}{2}$

For More Join: @IITJEE_Advanc

7. Choose the correct statement, relevant to the given circuit



i) There is no current just after key closed

ii) There is maximum current just after key is closed.

iii) There is maximum potential difference across capacitor just after closing key.

iv) There is maximum charge on capacitor after long time of closing switch

Choose the correct option relevant to above statement

A) ii, iv only B) i, iii, iv only C) ii, iii, iv only D) iii, iv only

8. The position of a particle moving on x-axis is given by $x(t) = A\sin\left(\frac{t}{t_0}\right) + Bt + \frac{C}{D+t}$, where t is time. The dimensions of $\frac{ABC}{D}$ is

A) L B) L^2T^{-1} C) L^3T^{-1} D) L^2T^{-2}

9. Match the List – I with List – II

Column – I		Column – II	
A)	The internal energy of an ideal gas system remains constant	I)	Isobaric process
B)	The process which undergoes infinitely slow	II)	Polytropic process
C)	Pressure of an ideal gas varies inversely with the nth power of volume of gas	III)	Quasistatic process
D)	Heat absorbed by gas partially increases internal energy of gas and rest is spent in doing work.	IV)	Isothermal process

Choose the correct answer from the _____ given below

A) $A \to IV; B \to III; C \to II; D \to I$

For More Join: @IITJEE_Advanc

B) $A \to I; B \to II; C \to III; D \to IV$

C) $A \to II; B \to III; C \to IV; D \to I$

D) $A \to III; B \to IV; C \to I; D \to II$

10. Consider a moving coil galvanometer (MCG)

i) The torsional constant in MCG has dimensions $\left[ML^2T^{-2}\right]$

ii) Increasing the current sensitivity may not necessarily increase the voltage sensitivity

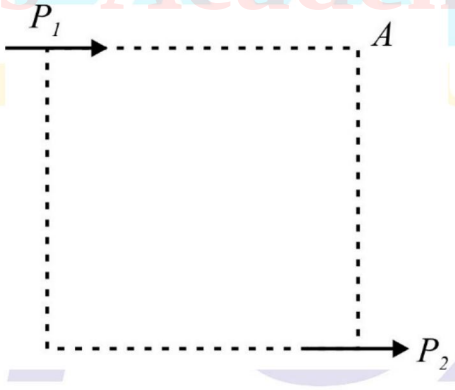
iii) If the no. of turns in coil is doubled then voltage sensitively doubles

iv) Current sensitivity of MCG is inversely proportional to number of turns in the coil

choose the correct answer from the options given below

A) i, ii only B) i, iv only C) ii, iv only D) i, ii, iv only

11. Two short dipoles are arranged at the diagonally opposite corners of a square of side ℓ, as shown in the diagram. It is observed that the resultant electric field at point A is zero then the value of $\left(\frac{p_1}{p_2}\right) =$



A) $\frac{1}{2}$ B) 1 C) 2 D) $\sqrt{2}$

For More Join: @IITJEE_Advanc

12. A lead bullet of mass 25 gm moving with speed v hits a wall at room temperature $\left(27^0 C\right)$ and melts down. The speed by which bullet is hitting the wall (melting temperature of lead = 600K, latent heat of fusion of lead $=2.5\times10^4 J/kg$ and specific heat capacity of lead $=125 J/kg-k$, Assume all the heat energy released in the collision is absorbed only by the bullet)

A) 338 m/s B) 354 m/s C) 366 m/s D) 375 m/s

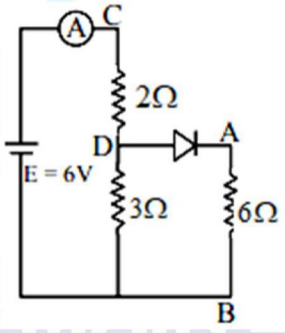
13. A ray of light incident on a glass slab with angle of incidence 60^0 emerges from another parallel side with lateral shift of 5 cm. If the thickness of the glass slab is $5\sqrt{3}\,cm$ then refractive index of slab is

A) $\sqrt{2}$ B) $\sqrt{3}$ C) $\sqrt{\frac{3}{2}}$ D) 2

14. A solid sphere and disc of same radius rolling down is an inclined plane from the same point. Then the ratio of their speed at the bottom of the plane, respectively

A) $\sqrt{\frac{14}{15}}$ B) $\sqrt{\frac{15}{14}}$ C) $\sqrt{\frac{9}{5}}$ D) $\sqrt{\frac{5}{9}}$

15. Refer to the circuit diagram given in the figure. Select correct observation (the diode is ideal)



i) Total resistance of circuit is 4Ω

ii) Ammeter reading is 1.5A

iii) Potential difference across AB is 3 volts

iv) Current across 3Ω resistance 1 Amp

A) i, ii, iii only B) i, ii, iii, iv only C) ii, iii, iv only D) i, ii, iv only

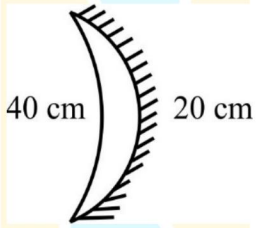
For More Join: @IITJEE_Advanc

16. The electric flux is defined as $\phi = A\rho + B\sigma + C\lambda$ where ρ, σ and λ are volume charge density, surface charge density and linear charge density respectively then $\left(\frac{A}{BC}\right)$ represent

A) Electric field B) Charge

C) Electric permittivity D) Area

17. A concavo – convex lens made of glass of refractive index 1.5 have radius of curvature 20 cm and 40 cm. If convex surface of lens is silvered then find the position of object in front of the lens so that its image coincide with it



A) 8 cm B) 16 cm C) 4 cm D) 12 cm

18. The electric field of an electromagnetic wave in free space is $\vec{E} = 47\cos\left[9\times10^{6}t - 6\times10^{-3}\left(3x+4y\right)\right]\left(4i-3\hat{j}\right)$. The associated magnetic field in tesla is

A) $B = -\frac{47}{3\times10^{8}}\cos\left[9\times10^{6}t - 6\times10^{-3}\left(3x+4y\right)\right]\hat{k}$

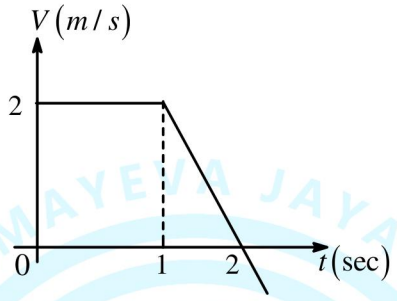
B) $\vec{B} = -\frac{47}{3\times10^{8}}\cos\left[9\times10^{6}t - 6\times10^{-3}\left(3x+4y\right)\right]\left(5\hat{k}\right)$

C) $\vec{B} = \frac{47}{3\times10^{8}}\cos\left[9\times10^{6}t - 6\times10^{-3}\left(3x+4y\right)\right]\left(5\hat{k}\right)$

D) $\vec{B} = -\frac{47}{3\times10^{9}}\cos\left[9\times10^{6}t - 6\times10^{-3}\left(3x+4y\right)\right]\left(\hat{k}\right)$

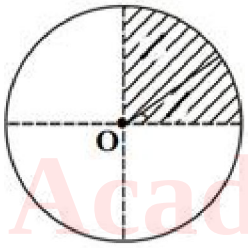
For More Join: @IITJEE_Advanc

19. A particle moves in a straight line. It's velocity time $(v-t)$ graph is as shown in figure. The distance travelled by particle in first two seconds is



A) 1m B) 2m C) 3m D) 4m

20. The disc of mass M radius R with uniform mass density is shown in figure. The distance of centre of mass of quarter disc (the shaded area) from the centre (O) of disc is



A) $\frac{2\sqrt{2}R}{3\pi}$ B) $\frac{4\sqrt{2}R}{3\pi}$ C) $\frac{4R}{3\pi}$ D) $\frac{2R}{3\pi}$

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

21. A positive ion A and a negative ion B has charges $6.67\times10^{-19}C$ and $9.6\times10^{-10}C$ and masses $19.2\times10^{-27}kg$ and $9\times10^{-27}kg$ respectively. They are separated at a certain distance r. The ratio of magnitude of electrostatic force to gravitational force is $P\times10^{44}$, where the value of P is

$(\frac{1}{4\pi\in_0}=9\times10^9 Nm^2/c$ and $G=6.67\times10^{-11}Nm^2kg^{-2})$

For More Join: @IITJEE_Advanc

22. The position vector of a moving body at time 't' is given as $\vec{r} = \left(5t^2\hat{i} - 5t\hat{j}\right)m$. The angle between position vector and velocity vector at $t = 1\,\text{sec}$ is $\cos^{-1}\left(\frac{P}{\sqrt{10}}\right)$, the value of P is

23. An ideal gas initially at 0^0C temperature is compressed suddenly. The ratio of specific heat at constant pressure to that at constant volume is $\frac{3}{2}$. The change in temperature due to the process is 273K. The final volume becomes $\left(\frac{1}{n}\right)$ times the initial volume, the value of n is

24. A force $\vec{F} = xy\hat{i} + y^2\hat{j}$ act on a particle is and moves along the line $x + y = 10$ the work done by this force during displacement from (0,10m) to (2m, 8m) in joule

25. A coil having 100 turns and area of 0.001 m^2 is free to rotate about an axis; the coil is placed with its plane perpendicular to magnetic field of1.0 $\frac{Wb}{m^2}$. If the coil is rotated rapidly through an angle of 180^0, how much charge will flow through the coil (in coulomb) ? The resistance of the coil is 0.1Ω

Rankers Academy JEE EDUCATIONAL INSTITUTIONS
APEX JEE-MAIN 2026

For More Join: @IITJEE_Advanc

CHEMISTRY **MAX.MARKS: 100**

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.

Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

26. Which of the following elements belongs to 6th period & 8th group of modern periodic table?

A) Pt B) Ds C) Rb D) Os

27. An e^- in a $H-atom$ undergoes deexcitation from a higher energy state to a lower energy state. The wavelength of radiation emitted for that transition is equal to $\frac{36}{5R_H}\left(R_H = \text{Rydberg constnat}\right)$. The spectral line is___ (Given $R_H = 10^5/cm, h = 6.6\times10^{-34} J\,s, c = 3\times10^8 m/s$)

A) Paschen series $\infty \to 3$ B) Lyman series $\infty \to 1$

C) Balmer series $3 \to 2$ D) Balmer series $\infty \to 2$

28. Incorrect statements among the following:

A) Tendency to exhibit –3 oxidation state decreases down the group in 15th group.

B) In a chemical reaction HNO_2 disproportionate to give HNO_3 and N_2O_5.

C) P and As can form $d\pi - P\pi$ bonds also with transition metals when their compounds like $P\left(C_2H_5\right)_3$ and As $\left(C_2H_5\right)_3$ act as ligands.

D) SO_2 can acts as both oxidizing and reducing agent in chemical reactions

29. $PtCl_4.6H_2O$ can exists as hydrated complex. 1.0m aqueous solution of this complex has depression in freezing point as 3.72^0C. Assume 100% of ionization of this complex. (k_f of water is $1.86K\ kg/mol$). Then the correct formula of the complex is

A) $\left[Pt\left(H_2O\right)_6\right]\ Cl_4$ B) $\left[Pt\left(H_2O\right)_4 Cl_2\right]Cl_2.2H_2O$

C) $\left[Pt\left(H_2O\right)_3 Cl_3\right]Cl.3H_2O$ D) $\left[Pt\left(H_2O\right)_2 Cl_4\right].4H_2O$

For More Join: @IITJEE_Advanc

30. $AO_4^{-2} \xrightarrow[n_1=3]{E_1^0=3V} A^{3+} \xrightarrow[n_2=1]{E_2^0=-0.6V} A^{2+} \xrightarrow[n_3=2]{E_3^0=-0.4V} A$ In the above diagram, standard electrode potentials are given in volts. Then value of $E_{AO_4^{-2}|A^{2+}}$ is

A) 1.7V B) 1.2 V C) 2.1 V D) 1.4 V

31. Match the following

Test		Reagent	
1)	Lucas test	a)	$C_6H_5SO_2Cl$
2)	Hinsberg test	b)	$Alc\ KOH + CHCl_3$
3)	Iso cynide test	c)	Anhyd $ZnCl_2 + Conc.HCl$
4)	Molisch test	d)	$Alc\,\alpha - \text{naphthol} + Conc.H_2SO_4$

A) $1-c, 2-a, 3-b, 4-d$ B) $1-a, 2-c, 3-b, 4-d$

C) $1-c, 2-a, 3-d, 4-b$ D) $1-c, 2-b, 3-a, 4-d$

32. **Statement-I :** Fructose does not contain $-CHO$ group and doesn't reduce Tollens reagent

Statement-II: In the presence of base fructose undergoes rearrangement to give glucose

In the light of above statements, choose the correct answer from the options given below:

A) Statement-I is false but statement-II is true

B) Both statement-I and statement-II are true

C) Both statement-I and statement-II are false

D) Statement-I is true but statement-II is false

33. 1.5 milli moles of N_2O is left after removing 6×10^{20} molecules from $x\ mg$ of sample . The mass of N_2O taken in initially in milligrams is

A) 110 B) 210 C) 322 D) 222

34. Ice at -3^0C is heated to water vapor at a temperature of 117^0C and 1.0 atm pressure. The entropy change associated with this process can be obtained from

A) $\int_{270}^{273} C_{Pm}.dT + \frac{\Delta H_f}{273} + \frac{\Delta H_v}{373}$

For More Join: @IITJEE_Advanc

B) $\int_{270}^{273}\frac{C_{Pm}.dT}{T}+\frac{\Delta H_f}{273}+\int_{273}^{373}\frac{C_{pm}.dT}{T}+\frac{\Delta H_v}{373}+\int_{373}^{390}\frac{C_{pm}.dT}{T}$

C) $\int_{270}^{273}C_{Pm}.dT+\frac{q_{rev}}{T}$

D) $\int_{270}^{273}C_{Pm}.dT+\frac{\Delta H_f}{273}+\frac{\Delta H_v}{373}+\int_{273}^{273}C_{Pm}.dT$

35. $[Fe(CN)_6]^{3-}$ ion has magnetic moment of 1.73 BM then $[Fe(H_2O)_6]^{3+}$ has a magnetic moment of

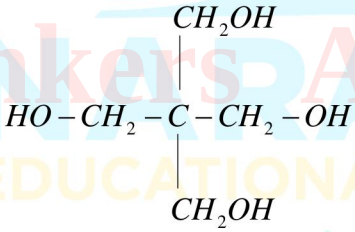
A) 1.73 BM B) 5.91 BM C) 2.82 BM D) 0 BM

36. Complex showing facial and meridional isomerism

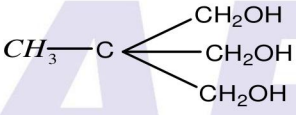
A) $[Pt(NH_3)_3Cl]Cl$ B) $[Zn(NH_3)_2Cl_2]$

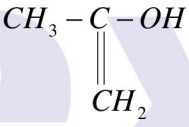
C) $K_3[Fe(C_2O_4)_3]$ D) $[Co(NH_3)_3Cl_3]$

37. $CH_3-CHO+H-CHO \xrightarrow[OH^- \text{ Reflux}]{excess\ H-CHO}$ A, then A is

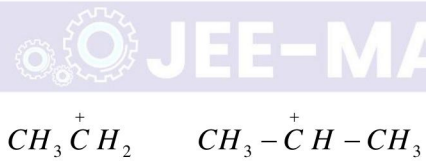
A) 

B) CH_3-CH_2OH

C) 

D) 

38. Compare the stability of carbocations



I II III

A) $III > II > I$ B) $II > III > I$ C) $III > I > II$ D) $II > I > III$

For More Join: @IITJEE_Advanc

39. Number of products formed in the monochlorination of ethyl cyclohexane (Ignore stereo isomers)

A) 6 B) 8 C) 5 D) 9

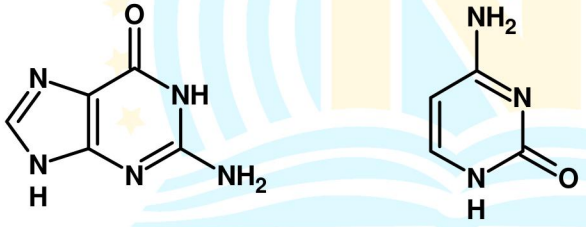
40. What amount of Br_2 in grams will be required to convert 9.4g of phenol into 2,4,6- Tribromo phenol (Given molar mass in g mol^{-1} of $C, H, O \& Br$ are 12,1,16 & 80)

A) 48 B) 96 C) 192 D) 20.44g

41. Which of the following chloride does not respond to the chromyl chloride test?

A) NaCl B) $CaCl_2$ C) $ZnCl_2$ D) Hg_2Cl_2

42.



Guanine **Cytosine**

In a DNA double - helix, guanine and cytosine bases, shown above, are paired together by

A) covalent bonds B) hydrogen bonds

C) peptide bonds D) Ionic bond

43. In a given aqueous solution $\left[A^{3+}\right] = \left[B^{2+}\right] = 1.0M$. When NH_4OH is added to the same solution which of the following is observed.

(Given $K_{SP}\left[A(OH)_3\right] = 27 \times 10^{-18}$; $K_{SP}\left[B(OH)_2\right] = 9 \times 10^{-10}$ at 298k)

A) $A(OH)_3$ will precipitate before $B(OH)_2$

B) $A(OH)_3$ & $B(OH)_2$ will precipitate together

C) $B(OH)_2$ will precipitate before $A(OH)_3$

D) Both $A(OH)_3$ & $B(OH)_2$ do not show precipitation

For More Join: @IITJEE_Advanc

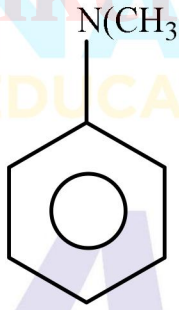
44. Identify the correct change from reactant to product

$BF_3 + NH_3 \rightarrow [NH_3 \rightarrow BF_3]$

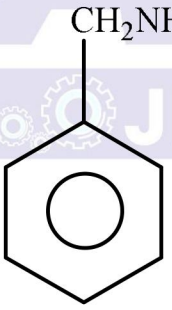
(X) (Y) $[Y - X]$

	Hybridization	Shape
A)	$X: sp^2 - sp^3$ $Y: sp^3 - sp^3$	X: Trigonal planar to tetrahedral Y: Trigonal pyramidal to tetrahedral
B)	$X: sp^3 - sp^3$ $Y: sp^2 - sp^3$	X: No change Y: square planar to tetrahedral
C)	$X: sp^3 - sp^2$ $Y: sp^2 - sp^3$	X: No change Y: Trigonal pyramidal to tetrahedral
D)	$X: sp^3 - sp^2$ $Y: sp^2 - sp^3$	X: Tetrahedral to trigonal planar Y: Trigonal pyramidal to square planar

45. Which among the following react with Hinsberg reagent

A) 

B) $CH_3 - NH - CH_3$ C) CH_3NH_2

D) 

E) $(C_6H_5)_3N$

A) A,C only B) C,D only C) B,C,D only D) A,C,E only

For More Join: @IITJEE_Advanc

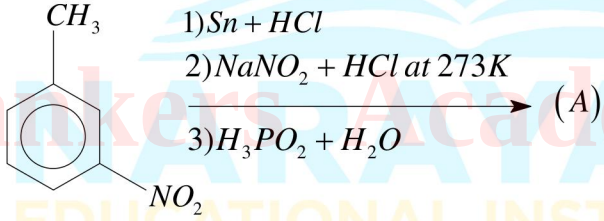
SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

46. The specific conductivity of 0.001 M solution of certain weak monoprotic acid $(\alpha << 1)$ is 4.95×10^{-5} s cm^{-1}. If λ_m^0 for the acid is 495 S cm^2 mol^{-1} then pH of the given solution is_____

47. The volume in mL, of $0.02M\ K_2Cr_2O_7$ solution required to react with $0.288\,g$ of ferrous oxalate (FeC_2O_4) in acidic medium is____. (Molar mass of $Fe = 56\,g\,mol^{-1}$)

48. Consider the following reactions to produce major product



Molar mass of product (A) _____ $g\,mol^-$

(Given molar mass in $gmol^-$ $C-12$ $H-0,16, N-14, P-38$)

49. The reaction $2N_2O_{5(g)} \rightarrow 2N_2O_{4(g)} + O_{2(g)}$ is first order in N_2O_5 at constant volume. The following data is obtained for the reaction.

S.NO	TIME (sec)	TOTAL PRESSURE (atm)
1.	0	1.2
2	60	X

The value of X = _____ $\times10^{-2}\,atm$

Given rate constant of the reaction is $2.303\times10^{-2}\,s^{-1}$ *and* $10^{0.6} = 4$

For More Join: @IITJEE_Advanc

50. If at a certain temperature, the following equilibrium is established,

$$CO(g)+NO_2(g)\rightleftharpoons CO_2(g)+NO(g)$$

One mole of each of the four gases is mixed in one litre container and the reaction is allowed to reach equilibrium state. When excess of baryta water is added to the equilibrium mixture, the weight of white precipitate obtained is 236.4 g. The equilibrium constant, K_c of the reaction is (at.wt of Ba=137)

MATHEMATICS MAX.MARKS: 100

SECTION – I
(SINGLE CORRECT ANSWER TYPE)

This section contains 20 multiple choice questions. Each question has 4 options (1), (2), (3) and (4) for its answer, out of which **ONLY ONE** option can be correct.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

51. $\int_0^{\infty} f\left(x+\frac{1}{x}\right).\frac{\ln x}{x}dx$:

A) is equal to zero B) is equal to one

C) is equal to $\frac{1}{2}$ D) can not be evaluated

52. The primitive of $\frac{1}{(x-4)^{3/2}(3-x)^{1/2}}$ is (where 'C' is integration constant):

A) $\left(\frac{3-x}{x-4}\right)^{1/2}+C$ B) $\frac{3}{4}\left(\frac{3-x}{x-4}\right)^{1/2}+C$ C) $\left(\frac{x-4}{3-x}\right)^{1/2}+C$ D) $2\sqrt{\frac{3-x}{x-4}}+C$

53. If $f(x)=\begin{cases} 3\left(1+|\tan x|\right)^{\frac{\alpha}{|\tan x|}}; \frac{-1}{2}<x<0 \\ \beta; \quad x=0 \\ 3\left(1+\left|\frac{\sin x}{3}\right|\right)^{\frac{6}{|\sin x|}}; 0<x<\frac{2}{3} \end{cases}$

is a continuous function at $x=0$, then the ordered pair (α,β) is equal to:

A) $\left(2,e^2\right)$ B) $\left(2,\frac{2}{e^2}\right)$ C) $\left(2,3e^2\right)$ D) $\left(2,\frac{3}{e^2}\right)$

For More Join: @IITJEE_Advanc

54. If A and B are points on the parabola $y^2 = 4ax$ with vertex O such that OA perpendicular to OB and having lengths r_1 and r_2 respectively, then the value of $\frac{r_1^{4/3} r_2^{4/3}}{r_1^{2/3} + r_2^{2/3}}$ is:

A) $16a^2$ B) a^2 C) $4a$ D) none of these

55. The solution of differential equation $e^{-x}(y+1)dy + (\cos^2 x - \sin 2x)ydx = 0$ subject to the condition $y = 1$ when $x = 0$, is :

A) $(y+1) + e^x.\cos^2 x = 3$ B) $\ln(y+1) + e^x.\cos^2 x = 1 + \ln 2$

C) $y + \ln y = e^x.\cos^2 x$ D) $y + \ln y + e^x.\cos^2 x = 2$

56. Domain of the function defined by $f(x) = \frac{1}{\sqrt{\sin^{-1} x}} + \log_{10}(\cos^{-1} x)$ is given by:

A) $[0,1]$ B) $(0,1]$ C) $[0,1)$ D) $(0,1)$

57. In the isosceles triangle $ABC, |\overrightarrow{AB}| = |\overrightarrow{BC}| = 8$, a point E divides AB internally in the ratio 1:3 then the cosine of the angle between $\overrightarrow{CE}$ and $\overrightarrow{CA}$ is : (where $|\overrightarrow{CA}| = 12$)

A) $-\frac{3\sqrt{7}}{8}$ B) $\frac{3\sqrt{8}}{17}$ C) $\frac{3\sqrt{7}}{8}$ D) $-\frac{3\sqrt{8}}{17}$

58. The sum of first 16 terms an A.P is 120. The second term of another A.P. whose first term is $\frac{2}{15}$ times of first term of given A.P. and common difference is same as common difference of given A.P. is :

A) 15 B) 0 C) 1 D) -15

59. The value of α for which the lines $\frac{x-2}{1} = \frac{y-9}{2} = \frac{z-13}{3}$ and $\frac{x-\alpha}{-1} = \frac{y-7}{2} = \frac{z+2}{-3}$ intersect, is

A) – 5 B) – 2 C) 5 D) – 3

For More Join: @IITJEE_Advanc

60. There exist a complex number z satisfying both $|z-3| \le 4$ and $|z+1| = a$ then the exhaustive set of value of a is____

A) $0 \le a \le 2$ B) $0 \le a \le 8$ C) $2 \le a \le 8$ D) $2 \le a \le 4$

61. Let $R = \{(a,b),(a,c),(b,b)\}$ be a relation defined on the set $\{a,b,c,d,e\}$. Then the minimum number of elements needed to be added in R so that R becomes an equivalence relation is

A) 10 B) 8 C) 11 D) 9

62. Six persons A,B,C,D,E and F are to be seated at a circular table. The number of ways this can be done if A must have either B or C on his right and B must have either C or D on his right is

A) 36 B) 12 C) 24 D) 18

63. In the ΔABC, the coordinates of B are (0,0), $AB = 2, \angle ABC = \pi/3$. If the middle point of BC has coordinates (2,0), then the centroid of ΔABC is :

A) $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ B) $\left(\frac{5}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ C) $\left(\frac{4+\sqrt{3}}{3}, \frac{1}{3}\right)$ D) $\left(\frac{5}{3}, \frac{1}{\sqrt{3}}\right)$

64. The value of α for which $\alpha x + \sec^{-1}\sqrt{2x^2 - x^4} + \cot^{-1}\left(\sqrt{x^2(2-x^2)} - 1\right) = 0$:

A) $\frac{\pi}{4}$ B) $-\frac{\pi}{2}$ C) $\frac{2}{\pi}$ D) $-\frac{2}{\pi}$

65. If $\sin x + \sin y + \sin z + \sin w = -4$, then the value of $\sin^{400} x + \operatorname{cosec}^{300} y + \sin^{200} z + \operatorname{cosec}^{100} w$ is:

A) $\sin^{400} x . \sin^{300} y . \sin^{200} z + \sin^{100} w$ B) $\sin x . \sin y . \sin z . \sin w$

C) 4 D) 3

66. For a statistical data $x_1, x_2, ..., x_{10}$ of 10 values, a student obtained the mean as 5.5 and $\sum_{i=1}^{10} x_i^2 = 371$. He later found that he had noted two values in the data incorrectly as 4 and 5, instead of the correct values 6 and 8 respectively. The variance of the corrected data is

A) 9 B) 5 C) 7 D) 4

For More Join: @IITJEE_Advanc

67. Let the position vectors of three vertices of a triangle be $4\vec{p}+\vec{q}-3\vec{r}$, $-5\vec{p}+\vec{q}+2\vec{r}$ and $2\vec{p}-\vec{q}+2\vec{r}$. If the position vectors of the orthocenter and the circumcenter of the triangle are $\alpha\vec{p}+\beta\vec{q}+\gamma\vec{r}$ and $\frac{\vec{p}+\vec{q}+\vec{r}}{4}$ respectively, then $\alpha+2\beta+5\gamma$ is equal to

A) 3 B) 4 C) 1 D) 6

68. Matrix A satisfies $A^2=2A-I$ where I is the identity matrix, then for $n\geq 2, A^n$ is equal to $(n\in N)$:

A) $nA-I$ B) $2^{n-1}A-(n-1)I$ C) $nA-(n-1)I$ D) $2^{n-1}A-I$

69. If system of equations $kx+2y-z=2, (k-1)x+ky+z=1, x+(k-1)y+kz=3$ has only one solution, then number of possible real value(s) of k is:

A) 0 B) 1 C) 2 D) infinite

70. A person have 3 normal dice, one red, one blue and one green and he rolls all three simultaneously. Let P be the probability that the sum of the numbers on red and blue dice is equal to the number on the green die. If P is written in lowest terms as $\frac{a}{b}$, then the value of $(a+b)$ equals

A) 79 B) 77 C) 61 D) 57

SECTION – II
(Numerical Value Answer Type)

This section contains 5 questions. The answer to each question is a Numerical value. If the Answer in the decimals, **Round off to the nearest Integer only**. Have to **answer 5 questions** and question will be evaluated according to the following marking scheme.
Marking scheme: +4 for correct answer, 0 if not attempted and -1 if not correct.

71. The area (in square units) enclosed by $|y|-|x|=1$ and $x^2+y^2=1$ is_____

72. If $(9+\sqrt{80})^n=I+f$ where I,n are positive integers and $0<f<1$, then the value of $(I+f)(1-f)$=_____

For More Join: @IITJEE_Advanc

73. If the eccentricity of the hyperbola $x^2 - y^2\sec^2\alpha = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\alpha + y^2 = 25$, then value of acute angle α in degrees is _____

74. The sum of real roots of the equation of $x^2 + \frac{x^2}{(x+1)^2} = 3$ is_____

75. Minimum number of real roots of the equation $(10px^2 - qx + r)(px^2 - qx - 5r)$ $(5px^2 - qx - r) = 0, (qpr \neq 0)$ is ____

Rankers Academy JEE
NARAYANA EDUCATIONAL INSTITUTIONS
APEX JEE-MAIN 2026

For More Join: @IITJEE_Advanc